NSBM SportsHub – R&D and Design Document

Oct 17, 2026 · @Jassim , @Shahilan

1. Document control

SportsHub is a PHP/MySQL web app that lets NSBM sports clubs run memberships, training, fixtures, tournaments and facility bookings, with separate Admin and Student panels. This document is the team's single reference for requirements, design, database, build plan and submission.

Item	Detail
Project title	NSBM SportsHub – Sports Club Management System
Module	Web and Mobile Application Development
Batch	26.1 IFSL
Faculty	Faculty of Computing, NSBM Green University
Stack	HTML, CSS, JavaScript (Bootstrap 5), PHP 8, MySQL 8
Submission deadline	Nov 12

Team and roles

Fill in names and contact details. The split below gives each member a visible, separate slice of code so the Git history shows everyone's contribution (a marked requirement).

Member	Role	Owns in the code
[Mukilan] – Team Leader	Project lead, backend lead	Auth, sessions, roles, DB connection layer, deployment
	Database and reports	Schema, seed data, admin reports, standings logic
	Admin panel developer	Clubs, memberships, sessions, matches, tournaments CRUD
	Student panel developer	Browse/join clubs, registrations, bookings, history
	UI/UX and QA	Layout, responsive CSS, JS validation, testing, report and video

Fewer than five members? Merge "UI/UX and QA" into the two panel roles.

2. Introduction and problem background

University sports clubs typically run on WhatsApp groups, paper sign-up sheets and spreadsheets kept by one club secretary. SportsHub replaces that with one shared system.

Problems the system addresses (validate these with 3–5 club members or captains before finalising – a short Google Form is enough and gives the report real R&D evidence):

- **Scattered membership records** – no single list of who belongs to which club, or who was approved.
- **Schedules lost in chat** – training times and fixture changes get buried in group messages.
- **Double-booked facilities** – two clubs ask for the same court or ground at the same time with no conflict check.
- **Manual results and standings** – scores are recorded by hand and league tables are recalculated by hand.
- **No participation evidence** – students cannot prove attendance or matches played (useful for CVs and awards), and staff cannot report on engagement.

SportsHub gives administrators one panel to manage clubs, schedules, tournaments, results and bookings, and gives students one place to join clubs, see what is on, register, book equipment and track their own history.

3. Objectives and scope

Objectives

1. Deliver a secure, role-based web app with separate Admin and Student panels.
2. Implement full CRUD for clubs, memberships, training sessions, matches, tournaments, facilities, equipment and bookings.
3. Prevent facility double-booking through a server-side time-overlap check.
4. Calculate tournament standings automatically from recorded results.
5. Give admins participation and performance reports with CSV export.
6. Give every student a personal participation history.
7. Be responsive from 360 px phones to desktops, and deploy live on a free PHP/MySQL host.

In scope

- Registration, login, logout, password hashing, role-based access (admin, student)
- Clubs, membership requests and approvals
- Training sessions and matches (scheduling, venues, results)
- Tournaments, team entries, fixtures, standings
- Facilities and equipment catalogue, booking requests and approvals
- Attendance marking and personal history
- Search and filtering on every list page
- Reports (participation, performance, bookings) with CSV export
- In-app notifications and announcements

Out of scope

- Online payments for membership fees (a "fee paid" flag only)
- Native Android/iOS apps (the responsive site covers mobile)
- Live score streaming and video

- Email/SMS delivery (free hosts usually block outgoing mail; in-app notifications instead)

4. Research: comparable systems

No existing product combines club membership, fixtures, tournaments and facility booking for a single university, so SportsHub borrows proven ideas from three tools and adds a university-specific admin layer.

System	What it does well	Gap for NSBM	What SportsHub adopts
Spond	Groups as the core unit; events for training and matches, including repeating events; attendance responses tracked per event; a separate web panel for club-wide admins	Built around one club; no tournaments, standings or venue booking; not tied to student IDs	Club = group model, recurring training sessions, RSVP and attendance per session, separate admin panel
TeamSnap	Players mark availability for each game or practice; coaches review past availability as an attendance record; availability can be locked shortly before an event	Commercial, aimed at youth teams and parents; advanced features on paid plans	Going / not going RSVP, registration cut-off before a match, attendance history per student
Skedda	Courts set up as bookable spaces; self-service booking enforced by rules such as quotas, buffer time and advance-notice limits	Booking only; nothing about clubs, members or results	Facility slots with conflict checking, max hours per booking, minimum advance notice, admin approval
Current NSBM practice (WhatsApp + Google Sheets)	Free and familiar	No approvals, no conflict check, no history, data owned by one person	Nothing; this is what SportsHub replaces

Gap SportsHub fills: one system where a student's identity links membership, attendance, match participation, tournament entries and bookings, so reports and personal history come for free from the same database.

For the report, add your own primary research: a 5-question survey to club members (how they hear about training, how often bookings clash, whether they want a participation record). Even 15–20 responses strengthen the R&D section.

5. User roles and permissions

Two roles are required by the brief: **admin** and **student**. A third, optional **coach/captain** role (a student flagged as captain of a club) is an easy innovation mark: they can mark attendance for their own club only.

Action	Guest	Student	Captain (optional)	Admin
View home page, club list, public fixtures and results	Yes	Yes	Yes	Yes
Register / login	Register	Login	Login	Login
Request to join a club, leave a club	–	Yes	Yes	–
Approve or reject membership requests	–	–	Own club	All clubs
Create, edit, delete clubs	–	–	–	Yes
Schedule training sessions and matches	–	–	Own club (training)	Yes
RSVP to training, register for matches and tournaments	–	Yes	Yes	–
Mark attendance	–	–	Own club	Yes
Create tournaments, generate fixtures, record results	–	–	–	Yes
Request a facility or equipment booking	–	Yes	Yes	Yes (direct)
Approve or reject bookings	–	–	–	Yes
Manage facilities, equipment, sports, users	–	–	–	Yes
View personal participation history	–	Own	Own	Any student
View reports, export CSV	–	–	–	Yes

Every admin page starts with a `require_role('admin')` check; every student page with `require_role('student')`. A captain is a student with `club_members.member_role = 'captain'`, checked per club.

6. Technology stack and hosting

Use plain PHP 8 with PDO, Bootstrap 5 and MySQL, and host on InfinityFree. Avoid frameworks such as Laravel: the brief marks PHP skills directly, free hosts make Composer deployment awkward, and every member can read plain PHP.

Layer	Choice	Why

Layer	Choice	Why
Markup and styling	HTML5, CSS3, Bootstrap 5.3 (CDN), Bootstrap Icons	Responsive grid, ready components (navbar, cards, modals, tables), allowed by the brief
Client scripting	Vanilla JavaScript; Chart.js for dashboard charts; FullCalendar (optional) for the schedule view	Light, no build step, works on free hosting
Server	PHP 8.x, procedural or simple classes, PDO	Mandatory in the brief; PDO gives prepared statements
Database	MySQL 8 locally (XAMPP), MySQL/MariaDB on the host, InnoDB tables, utf8mb4	Mandatory; InnoDB gives transactions and foreign keys
Local dev	XAMPP (Apache + MySQL + phpMyAdmin), VS Code	Free, matches the host's Apache/PHP setup
Version control	Git + GitHub (public repo), GitHub Desktop for less confident members	Commit history is marked
Design	Figma or pen-and-paper wireframes; draw.io or MySQL Workbench for the ER diagram	Screenshots for the report
Hosting	InfinityFree (free subdomain, free SSL)	Free, PHP 8.3 and MySQL, no credit card

Hosting constraints to design around

These come from InfinityFree's own forum and current reviews, and they shape the database design in section 10:

- **No views, triggers, stored procedures or temporary tables.** The database user only gets basic privileges such as SELECT, INSERT, UPDATE, DELETE, CREATE, ALTER, DROP and INDEX (InfinityFree forum). So standings, reports and conflict checks are plain SELECT queries run from PHP, never views or triggers.
- **Foreign keys:** older free plans only supported MyISAM, which silently ignores foreign keys; InnoDB support was added later (forum). Import the schema on the host in week 1 and run `SHOW CREATE TABLE club_members;` to confirm the constraints exist. If they do not, keep the same columns and enforce the rules in PHP.
- **No outgoing email:** PHP `mail()` is blocked (source), so use in-app notifications, not email verification or password-reset emails.
- **Connection limits:** too many simultaneous requests can trigger a temporary suspension, so do not load-test the live site.

- Plan specs as of 2026 listed by DeployHQ: 5 GB disk, PHP 8.3, free SSL. Keep uploaded images small (resize logos to under 300 KB).

Backup option if InfinityFree misbehaves close to the deadline: any other free cPanel-style PHP/MySQL host; the same files and SQL dump move across in about 30 minutes.

7. Local setup and deployment guide

Deploy a "hello world" with the login page in week 1; leaving the first deployment to the last week is the most common way groups lose the 15% for links.

Local setup (every member)

1. Install XAMPP; start Apache and MySQL.
2. `git clone` the repo into `C:\xampp\htdocs\sportshub`.
3. Open `http://localhost/phpmyadmin` → New → database `sportshub`, collation `utf8mb4_unicode_ci` → Import → `database/schema.sql`.
4. Copy `config/config.sample.php` to `config/config.php` and set `DB_HOST=localhost`, `DB_NAME=sportshub`, `DB_USER=root`, `DB_PASS=''`, `BASE_URL='http://localhost/sportshub'`.
5. Open `http://localhost/sportshub` and log in as `admin@sportshub.nsbm.lk` / `Password@123`.

Deploying to InfinityFree

1. Sign up at infinityfree.com, create a hosting account and pick a free subdomain (for example `nsbm-sportshub.rf.gd` or similar). A new subdomain can take a while before it opens.
2. In the control panel open **MySQL Databases**, create database `sportshub`. Note the four values the panel shows: MySQL host name, database name (prefixed with your account name), user name and password.
3. Open **phpMyAdmin** for that database → Import → `schema.sql`. Then run `SHOW CREATE TABLE club_members;` and check the `CONSTRAINT ... FOREIGN KEY` lines are there (section 8).
4. Upload the project into the `htdocs` folder with the online File Manager or FileZilla (FTP details are in the panel). Upload the contents of `sportshub/`, not the folder itself, so the site opens at the root.
5. Create `config/config.php` on the server with the host's database values and `BASE_URL = ''` (or the full https address).
6. Turn on the free SSL certificate in the panel, then force HTTPS with a small rule in `.htaccess`.
7. Log in as admin, **change the admin password**, and run the test plan.

Redeploying

- Upload only changed files after each Thursday merge.
- Database changes: write them as `ALTER TABLE` statements in `database/migrations/002_add_x.sql` and run them in phpMyAdmin. Do **not** re-import `schema.sql` on the live site; it starts with `DROP TABLE` and wipes all data.
- Export a live SQL backup from phpMyAdmin before every database change and before the video recording.

Demo accounts (seed data)

Role	Email	Password
Admin	admin@sportshub.nsbm.lk	Password@123
Student (football captain)	kavindu@students.nsbm.ac.lk	Password@123
Student	nethmi@students.nsbm.ac.lk	Password@123
Student	tharindu@students.nsbm.ac.lk	Password@123
Student (pending badminton request)	ayesha@students.nsbm.ac.lk	Password@123

Put these in the report and the README so the lecturer can test the live site without registering.

-